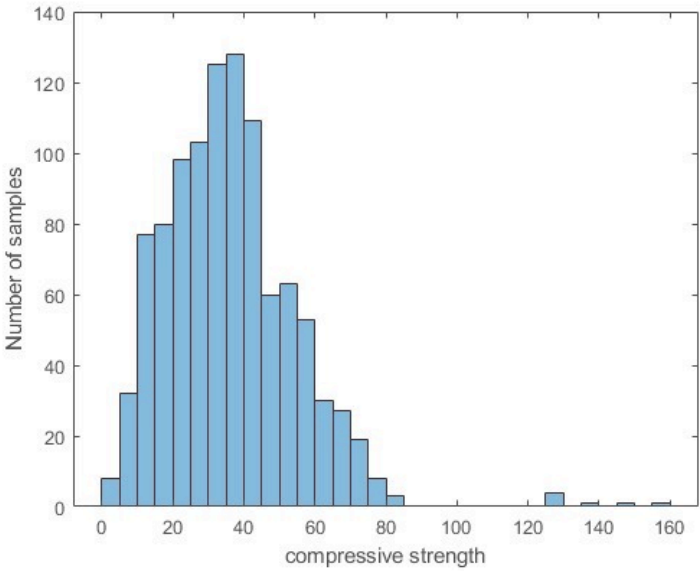


Outlier removal using the Z-score method

Concrete is the most important material in civil engineering. The concrete compressive strength is a highly nonlinear function of age and ingredients, such as cement content, water-to-cement ratio. The modified concrete compressive strength dataset [1] contains over 1,000 data instances that can be used to analyze the nonlinear relation between the factors and the resulting compressive strength.



[1] I-Cheng Yeh, "Modeling of strength of high performance concrete using artificial neural networks," Cement and Concrete Research, Vol. 28, No. 12, pp. 1797-1808 (1998)

As a civil engineer, you are analyzing the dataset (see reading materials), and believe that there are outliers in compressive strength that need to be removed before the dataset can be used for model training.

Please identify these outliers using the **Z-score method**.

* The compressive strength values can be imported using the code below.

```
T = readtable("Concrete_Data.xls", 'VariableNamingRule','preserve');
values = table2array(T(:,9));
```

** For the z-score methods, an appropriate z should often be $-3 \leq z \leq 3$.

Script

 Save

 Reset

 MATLAB Documentation (https://www.mathworks.com/help/)

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```
8 X = values;
9
10 % Calculating the z-score of every sample in the dataset; saving in z.
11 % Your code:
12 z = (values - mean(values)) / std(values);
13
14 % Plotting the distribution of z
15 histogram(z);
16
17 % Filtering out those samples using z-scores method
18 % Removing outliers
19 % Your code:
20 X = X(abs(z) <= 3);
21
22 % Visualize at last
23 histogram(X)
```

 Run Script



Tests: All Tests Passed

Submit



 Keep samples that are not outliers

 Outlier test #1

 Outlier test #2

Output

